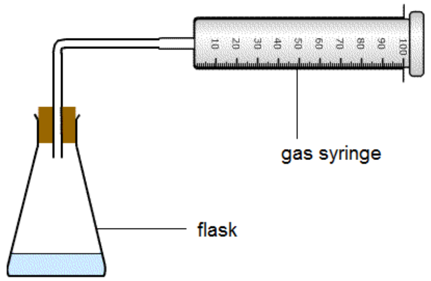
Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9.	(a)			 flask with stopper and delivery tube (1) method of collecting and measuring volume of oxygen (1) accept collection over water using measuring cylinder	2			2		2
	(b)	(i)		tangent drawn at t = 0 or t = 200 s (1) initial rate = 10-15 (cm^3s^{-1}) (1)		2		2	2	2
		(ii)		rate at t = 200s = 3-5 (cm^3s^{-1})		1		1	1	1
		(iii)		at t = 200 concentration lower / particles further apart (1) less frequent collisions / less chance of collisions, so rate less (1)	2			2		
		(iv)		2000 cm^3 oxygen = 2000/24500 mol = 0.0816 (1) number of moles H_2O_2 = 0.163 (1) concentration = 1.63 – must be correct to 3 sig figs (1)		2	1	3	1 1	

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)			any three of following for (1) each • measure volume of gas at a particular time / time how long it takes to produce a fixed volume • use the same amount of catalyst • use the same volume of the same concentration of H_2O_2 • better catalyst increases rate by more		1 1	1	3		3
				Question 9 total	**4**	**7**	**2**	**13**	**5**	**8**